

Sparse-View RGB + Estimated-Depth 3D Reconstruction

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Team Setup

Student information

- Nguyễn Ngọc Minh – 20235533
- Kiều Sơn Tùng – 20235571
- Đặng Thịnh Tường Minh – 20235528

Presentation goal

Present Run 30 as the validated RGB-D/source-depth reference and define the target RGB-only estimated-depth correction setting built from MV-DUSt3R+ candidates and the Run 37 fine-tuned depth estimator.

- 3D reconstruction is important because 3D representations help machines understand spatial structure, depth, and object relationships more accurately.
- Practical 3D capture often has only a small number of usable images.
- Classical high-quality pipelines depend on accurate camera poses or expensive scene-specific optimization.
- Sparse views increase ambiguity because cross-view correspondence is harder to establish reliably.
- A fast, generalizable method for unseen scenes would be useful for robotics, AR/VR, navigation, and scalable environment digitization.

Problem Statement

Core question

How can we reconstruct unseen indoor scenes accurately when only 2 to 5 sparse posed RGB-D views are available?

Key constraints

- Unseen scene at test time
- Sparse posed RGB-D views
- Known camera intrinsics/extrinsics
- No per-scene optimization

Expected outcome

- Robust geometry recovery
- Stable performance under view reduction
- Better occlusion and wrong-depth behavior
- Practical runtime per scene

Objectives

- ① Build a strong sparse-view MV-DUSt3R+ baseline for 2–5 views.
- ② Diagnose why RGB-only learned reliability, match filtering, and monodepth correction fail reconstruction-level gates.
- ③ Switch the final setting to posed RGB-D views with known camera intrinsics/extrinsics.
- ④ Validate Run 30 source-depth correction on overall, occlusion, and repeated/wrong-depth ambiguity subsets.

Final Hypothesis

Working hypothesis

Candidate deletion is too risky for sparse-view reconstruction. When input source depth is available, anchoring source rays to metric RGB-D geometry should correct wrong-depth candidates while preserving recall.

- Keep the useful MV-DUSt3R+ candidate set instead of over-filtering it
- Use source depth, poses, and intrinsics to correct 3D locations
- Test the result on occlusion and repeated/wrong-depth hard subsets

Our contribution

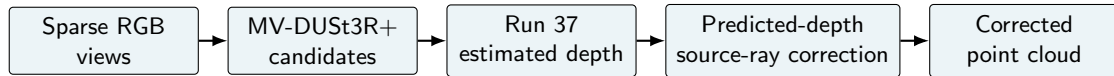
Run 30 validates the source-ray correction idea with true RGB-D source depth. The current target setting keeps MV-DUSt3R+ as the candidate generator and replaces true source depth with depth estimated from RGB by the Run 37 fine-tuned depth model.

- **Strong RGB-only baseline:** Run 11 improves B0 across 2–5 views using view selection and fixed confidence thresholds.
- **Negative RGB-only analysis:** Runs 22–29 show that proxy F1, candidate filtering, and generic monodepth do not solve reconstruction.
- **Validated RGB-D reference:** Run 30 uses input source depth maps with poses/intrinsics for source-ray correction and passes all gates.
- **Estimated-depth target:** Run 37 fine-tunes the RGB depth estimator; the next correction setting should use that estimated depth to align MV-DUSt3R+ candidates.

Target Reconstruction Setting

Inference contract

RGB-only sparse input images, known camera poses/intrinsics, MV-DUSt3R+ candidate geometry, and source depth estimated by the fine-tuned depth model.



Claim boundary

This target still uses known camera calibration and must pass the same reconstruction gates before replacing the Run 30 RGB-D reference result.

Predicted-Depth Generalization Track

- Goal: move from true source RGB-D depth toward RGB image input by estimating source depth from RGB, then using that estimate for correction.
- Run 35: evaluate Depth Anything V2 Metric Indoor Small with AbsRel, RMSE, MAE, $\delta_1/\delta_2/\delta_3$, and scale-aligned errors.
- Run 36: first predicted-depth correction attempt with the generic depth model; reconstruction gates did not pass.
- Run 37: fine-tune the depth estimator to provide a stronger estimated-depth correction anchor.
- Run 38: use the Run 37 `controlled_best` checkpoint for the full validation/test scene set by default, evaluating every discovered sparse-view group while policy selection remains based on the train split.
- True source depth is excluded from Run 36/38 method inference and residual gating; it remains evaluator/proxy-GT data only.
- Boundary: RGB-only with respect to source depth, but still uses known pose/intrinsics and is not the official MV-DUSt3R+ benchmark.

Runs 35–36 Results

- Run 35 test depth: AbsRel 0.1861, $\delta_1 = 0.7736$, raw MAE 0.3525 m, diagnostic scale-aligned MAE 0.1614 m.
- Run 36 selects global scale, $\tau_{\text{pred}} = 0.5$, and $\alpha = 0.25$.

Method	Val F-score	Test F-score
MV-DUSt3R+ RGB-only	0.1231	0.1744
Predicted-depth correction	0.1012	0.1465
Direct predicted depth	0.1747	0.1894
Run 30 true RGB-D	0.1753	0.2764

Outcome

Correction fails overall, occlusion, and ambiguity gates. Predicted depth is useful as a direct prior, but it does not replace Run 30 and does not solve the two RGB-only limits.

Contribution Details

1. Baseline that solves sparse-view instability

- Run 11 fixed thresholds and view selection beat B0 on 2, 3, 4, and 5 views.
- This solves the first limit but leaves occlusion and wrong-depth ambiguity as hard cases.

2. RGB-only diagnostic path

- OARH, RSDH, RAJAH, and monodepth corrections are tested through reconstruction-level gates.
- They expose why filtering hurts recall/completeness and why generic monodepth is not enough.

3. Run 30 source-depth correction

- Uses source RGB-D depth maps with known camera poses/intrinsics.
- Corrects source rays in metric geometry instead of deleting candidates.

Why RGB-only Failed

Transition narrative

Runs 22–29 showed that RGB-only filtering/correction was insufficient. Run 30 changes the inference contract to RGB-D and passes the overall, occlusion, and ambiguity gates.

- Proxy F1 did not transfer to reconstruction-level F-score.
- Filtering removed valid sparse-view geometry and hurt recall.
- All-candidate retention became a strong non-learned baseline.
- RGB-only monodepth did not recover metric/source depth well enough.

Proposed Method

- Primary method:
MV-DUS_t3R+
- Single-stage feed-forward multi-view reconstruction
- Final setting uses sparse posed RGB-D views
- Run 30 adds source-depth correction after MV-DUS_t3R+
- Our study focuses on robustness, hard cases, and runtime



Source: Tang et al., MV-DUS_t3R+, project page and paper

Architecture Comparison

DUST3R baseline

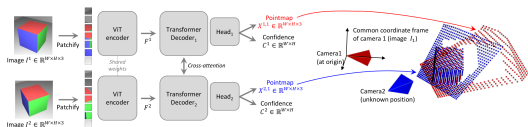
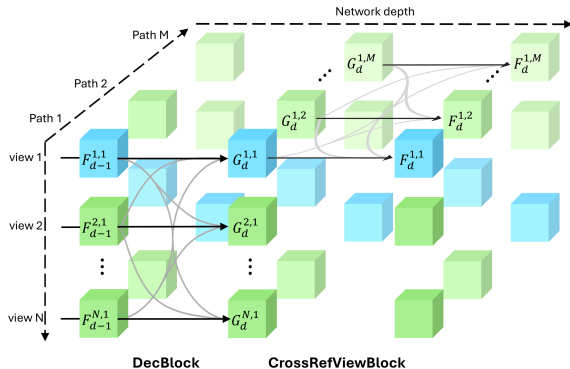


Figure 2. **Architecture of the network.** Two views of a scene (I^1, I^2) are first encoded in a Siamese manner with a shared ViT encoder. The resulting token representations F^1 and F^2 are then passed to two transformer decoders that constantly exchange information via cross-attention. Finally, two regression heads output the two corresponding pointmaps and associated confidence maps. Importantly, the two pointmaps are expressed in the same coordinate frame of the first image I^1 . The network is trained using a simple regression loss (Eq. (4))

The network then processes each view of them jointly in the two subsequent stages $\hat{V}^1, 1$ and $\hat{V}^2, 1$ obtained from Eq. (4)

Pairwise reasoning with cross-attention between two views

MV-DUST3R+ main method



Multi-path design with cross-reference-view information flow

Source: Left: Wang et al., DUST3R, CVPR 2024. Right: Tang et al., MV-DUST3R+, project page/paper

Why MV-DUS_t3R+ vs DUS_t3R?

Aspect	DUS _t 3R	MV-DUS _t 3R+
Processing style	Pairwise reconstruction	Joint multi-view reconstruction
Global consistency	Requires later alignment	Built into one forward pass
Sparse-view behavior	Can accumulate pairwise errors	More robust cue fusion
Runtime trend	Slower as views increase	Better scaling with more views
Role in project	Reference baseline	Main candidate method

- DUS_t3R is still a strong baseline, so the comparison is direct and meaningful.
- MV-DUS_t3R+ is the candidate generator for both the Run 30 RGB-D reference and the target estimated-depth correction step.

Quantitative Evidence from the Paper

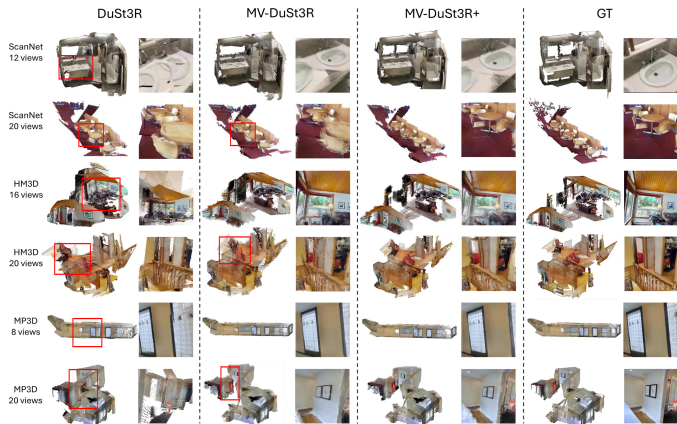
	Method	GO	HM3D			ScanNet			MP3D			Time (sec)
			ND ↓	DAc ↑	CD ↓	ND ↓	DAc ↑	CD ↓	ND ↓	DAc ↑	CD ↓	
4 views	Spann3R	×	37.1	0.0	225(184)	8.9	19.5	54.7(50.1)	42.7	0.0	248(202)	0.36
	DUST3R	✓	1.9	75.1	5.6(2.3)	1.3	89.8	4.0(0.4)	3.9	41.7	40.0(5.3)	2.42
	MV-DUST3R	×	1.1	92.2	2.0(1.1)	1.0	93.3	2.0(0.4)	2.5	62.4	25.3(4.1)	0.05
	MV-DUST3R+	×	1.0	95.2	1.5(0.9)	0.8	94.9	1.5(0.3)	2.2	68.0	19.9(3.4)	0.29
	MV-DUST3R _{oracle}	×	1.0	94.6	1.5(0.7)	0.8	95.5	1.3(0.3)	2.3	66.6	20.7(4.0)	-
	MV-DUST3R+ _{oracle}	×	0.9	96.5	1.4(0.7)	0.7	95.8	1.2(0.2)	2.1	70.6	17.9(3.3)	-
12 views	Spann3R	×	32.6	0.0	125(113)	9.1	16.3	36.6(31.2)	35.0	0.0	138(112)	1.34
	DUST3R	✓	3.9	30.7	18.1(3.4)	1.9	82.6	4.1(0.6)	6.6	12.0	49.6(8.3)	8.28
	MV-DUST3R	×	1.6	79.5	3.0(1.2)	1.4	86.8	2.3(0.8)	3.4	41.3	22.6(5.5)	0.15
	MV-DUST3R+	×	1.2	91.5	1.8(0.7)	1.2	88.4	1.8(0.7)	2.6	55.0	15.1(3.8)	0.89
	MV-DUST3R _{oracle}	×	1.3	88.8	1.8(0.9)	1.0	90.6	1.3(0.7)	2.9	51.3	16.4(4.0)	-
	MV-DUST3R+ _{oracle}	×	1.1	94.8	1.4(0.7)	1.0	90.9	1.3(0.5)	2.5	59.8	13.6(3.5)	-
24 views	Spann3R	×	41.7	0.0	139(121)	11.4	1.6	37.4(35.5)	46.6	0.0	151(121)	2.73
	DUST3R	✓	6.8	7.3	32.4(5.2)	2.4	72.6	5.1(1.0)	11.4	2.5	80.9(14.3)	27.21
	MV-DUST3R	×	3.4	36.7	10.0(3.5)	2.2	75.2	2.7(0.9)	6.3	12.2	38.6(13.9)	0.35
	MV-DUST3R+	×	2.1	64.5	3.9(2.0)	1.6	81.2	1.7(0.7)	4.3	26.7	22.0(5.9)	1.97
	MV-DUST3R _{oracle}	×	2.1	58.9	3.5(2.1)	1.4	82.9	1.4(0.7)	4.4	22.0	19.9(5.1)	-
	MV-DUST3R+ _{oracle}	×	1.8	77.9	2.6(1.3)	1.3	85.1	1.3(0.6)	3.6	33.1	15.1(4.4)	-

Table 2 MVS reconstruction results. Best results are marked in red. For clarity, metric ND is scaled up by 10×, DAc is in percent %, and CD is scaled by 100× with its median given in parentheses. Results of our **oracle** methods are obtained by manually selecting the best single reference view from reference view candidates to report a possibly achievable

Across multiple datasets and view counts, the highlighted rows show that MV-DUST3R+ improves reconstruction quality while keeping runtime practical.

Source: Tang et al., MV-DUST3R+, Table 2

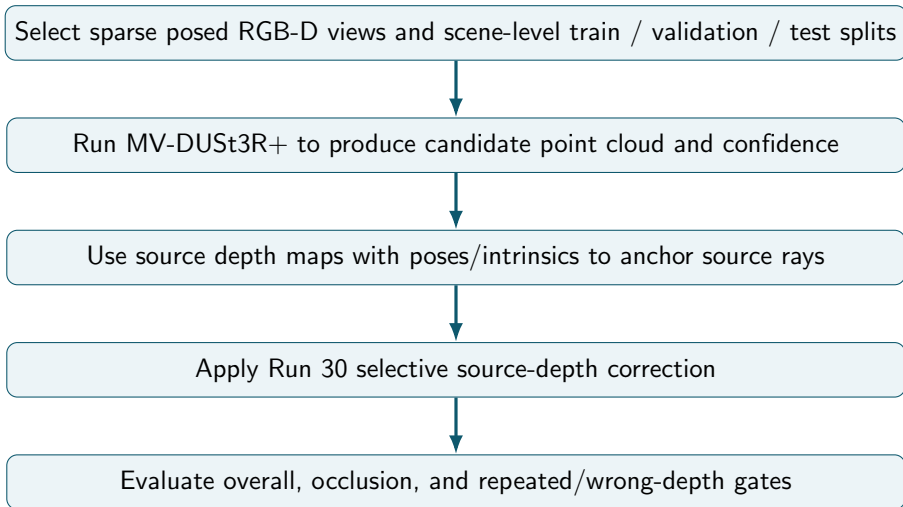
Qualitative Motivation



DUST3R often fails on repeated structures, while MV-DUST3R+ remains more stable in sparse-view scenes.

Source: Tang et al., MV-DUST3R+, qualitative results

Final Pipeline



Controlled factors

- Number of input views
- RGB-D source depth availability
- Scene-level train / validation / test split
- Fixed Run 30 policy on final test

Observed outcomes

- Reconstruction accuracy
- Completeness and consistency
- Runtime per scene
- Occlusion and repeated/wrong-depth failures

RGB-only Diagnostic Path

- ➊ First build a supervised label cache with per-view visibility, occlusion, floating/wrong-depth, and match labels.
- ➋ Mine occlusion-heavy, low-overlap, and hard-negative subsets before training the learned heads.
- ➌ Freeze MV-DUSt3R+ and train small OARH / RSDH heads from generated labels, excluding direct target-label leakage features.
- ➍ Runs 22–26 move the heads into reconstruction-level validation gates with fixed-confidence, top-k, and all-candidate baselines.
- ➎ Run 27 is self-contained and learns a bounded confidence residual from actual candidates, self-geometry support, and raw image-patch cues.
- ➏ Completed Run 27 still selects all candidates: learned filtering beats fixed confidence but loses completeness against full retention.
- ➐ These runs motivate switching the final inference contract to RGB-D.

RGB-D Source-Depth Correction

- ➊ Run 28 therefore preserves candidate count and learns source-ray 3D corrections from training depth and poses, using no depth at inference.
- ➋ Completed Run 28 still fails the gate, but its source-depth oracle shows large occlusion and ambiguity headroom.
- ➌ Run 29 approximates that oracle with RGB-only monocular depth aligned through the input poses and intrinsics.
- ➍ Completed Run 29 regresses both occlusion and ambiguity, so RGB-only monodepth is not enough.
- ➎ Run 30 makes source depth an explicit RGB-D input and gates full/selective source-ray correction policies.
- ➏ Selected policy: `rgbd_residual_ge_0.30`, residual mode, $\alpha = 1.0$, threshold $0.30m$.
- ➐ Final selected method: `rgbd_source_depth_selected`.
- ➑ Keep final test clean: no threshold tuning, no checkpoint selection on test scenes.

Controlled ScanNet-style RGB-D subset

The reported Runs 19–33 use posed indoor RGB-D scenes with source images, depth maps, and known camera intrinsics/extrinsics.

Scene-level split

- 18 train scenes
- 6 validation scenes
- 6 held-out test scenes
- No test scene used for training or policy selection

Sparse-view protocol

- 3, 4, and 5 selected views
- Hybrid and diversity-aware view groups
- Geometry metrics at a 0.05 m threshold
- Hard occlusion and ambiguity subsets

Evaluation Scope

What the current evidence supports

Held-out comparison of Run 30 against all-candidate and fixed-confidence baselines on the controlled ScanNet-style subset.

What is not claimed

The current results are not a full ScanNet++ evaluation and do not establish official-benchmark generality. A larger mesh or laser-scan benchmark remains future work.

- Run 11 remains the strongest RGB-only baseline.
- Runs 12–29 are diagnostic and negative evidence.
- Run 30 is the validated RGB-D/source-depth reference method.
- Run 37 prepares the estimated-depth source for the target RGB-only correction setting.
- Run 33 isolates MV-DUSt3R+ RGB-only outputs as the final backbone baseline.

Evaluation Plan

- Reconstruction F-score against available ground-truth geometry
- Accuracy, completeness, and Chamfer distance of point clouds
- Symmetric normalized distance and normalized DAc@0.2
- Sparse 2–5 view behavior across scene splits
- Limit-specific gates for occlusion and wrong-depth ambiguity
- Runtime per scene for practical inference

	Method	GO	HM3D			ScanNet			MP3D			Time (sec)
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Source: Tang et al., MV-DUS3R+, quantitative benchmark

Success criterion

Run 30 must beat the best non-learned baseline overall and improve both occlusion and ambiguity subsets by at least the gate margin.

Normalized Geometry Metrics

For each point cloud, remove translation and isotropic RMS scale:

$$\hat{X} = \frac{X - \bar{X}}{\sqrt{\frac{1}{|X|} \sum_{x \in X} \|x - \bar{X}\|_2^2}}.$$

Normalized distance

$$\text{ND}_{\text{sym}} = \frac{1}{2} (\text{mean } d(\hat{P}, \hat{G}) + \text{mean } d(\hat{G}, \hat{P}))$$

Lower is better; measures two-way normalized shape error.

DAc@0.2 normalized

$$\frac{|\{p \in \hat{P} : d(p, \hat{G}) \leq 0.2\}|}{|P|}$$

Higher is better; measures prediction-side normalized accuracy.

Scope

Auxiliary paper-style metrics only. They still use depth-derived proxy GT, and normalization does not remove rotation.

Run 34 Normalized Metric Results

Mean over three matched 3,500-point qualitative groups:

Method	Normalized distance	DAc@0.2
Run 30 RGB-D selected	0.0515	0.9841
Direct RGB-D backprojection	0.0926	0.8617
MV-DUSt3R+ RGB-only	0.1308	0.7967

Reading the result

Run 30 has the lowest normalized shape error and highest normalized prediction accuracy on this subset.

Claim boundary

This is a three-group qualitative comparison against depth-derived proxy GT. It complements, but does not replace, the full Run 30 F-score gates.

Final RGB-D Result

Run 30 gate decision

```
selected_method = rgbd_source_depth_selected; best baseline = all_candidates; gate margin = 0.005;  
pass_all_limits = 1.
```

Split/subset	All cand.	Fixed conf.	RGB-D selected	Delta vs all
Val overall	0.1194	0.1123	0.1753	+0.0559
Val occlusion	0.1064	0.0917	0.2522	+0.1458
Val ambiguity	0.1623	0.1416	0.3000	+0.1377
Test overall	0.1758	0.1662	0.2764	+0.1007
Test occlusion	0.2111	0.1866	0.3146	+0.1035
Test ambiguity	0.1621	0.1462	0.2948	+0.1327

Final wording

RGB-only learned extensions did not pass reconstruction-level gates. After switching the inference contract to RGB-D, Run 30 uses input source depth maps with known camera poses/intrinsics for source-ray correction and passes the overall, occlusion, and ambiguity gates on held-out scenes.

- Source depth anchors each source ray to metric geometry.
- Correction preserves candidate count, unlike filtering that hurts recall and completeness.
- Occlusion and repeated/wrong-depth cases improve because the method fixes geometry rather than discarding sparse-view evidence.

Run 31 Coverage Stress Test

Purpose

Validate the frozen Run 30 method over more sparse-view groups. Run 31 does not train, tune, or select a new reconstruction method.

Coverage

- 30 scenes
- 12 groups per scene
- 360 groups total
- 3, 4, and 5 views
- No dense-frame inference

Fixed evaluation

- Hybrid and diversity-aware groups
- Two frame variants per configuration
- Frozen `residual_ge_0.30` policy
- Paired F-score delta vs all candidates
- Scene-cluster bootstrap 95% CI

Current status

Submitted/pending result. No additional stability claim is made yet.

Direct RGB-D Backprojection Baseline

Run 32 question

How well does direct depth backprojection reconstruct the same sparse-view groups without MV-DUS_t3R+?

- Back-project valid depth with known intrinsics/poses and attach RGB.
- Apply fixed 2 cm voxel downsampling and cap at 3,500 points.
- Reuse Run 30 groups, metrics, and hard subsets; no policy tuning.

Subset	Direct voxel	Run 30	Direct sampled
Val overall	0.0874	0.1753	0.8500
Test overall	0.1359	0.2764	0.8666

Interpretation

Direct backprojection is not source-depth correction. The primary voxel direct baseline loses to Run 30, including occlusion and ambiguity subsets. The sampled diagnostic is very high because GT uses the same selected input depth maps; treat it as evaluator-circularity evidence, not an official mesh/laser-scan benchmark result.

Run 33: MV-DUSt3R+ Only RGB Baseline

Question

How much does MV-DUSt3R+ alone achieve using selected sparse RGB views, without source-depth correction or direct RGB-D backprojection?

- Reuses clean Run 30 `all_candidates` and `confidence_fixed_final` rows; no expensive rerun.
- Source-depth inference, correction, and direct RGB-D flags are false.
- Best RGB-only method is `mvdust3r_raw_all_candidates`.

Split/subset	RGB-only	Run 30	Delta
Val overall	0.1194	0.1753	+0.0559
Val occlusion	0.1064	0.2522	+0.1458
Val ambiguity	0.1623	0.3000	+0.1377
Test overall	0.1758	0.2764	+0.1007
Test occlusion	0.2111	0.3146	+0.1035
Test ambiguity	0.1621	0.2948	+0.1327

Interpretation

Run 30 adds value over MV-DUSt3R+ RGB-only outputs. The final claim remains RGB-D/source-depth; RGB-only stays a baseline and negative-evidence setting.

Run 37 To Estimated-Depth Correction

Why Run 37 exists

Run 36 showed that generic predicted depth was not accurate enough for source-ray correction. Run 37 fine-tunes the depth estimator on the available RGB-D frame pool so predicted depth can become a stronger correction anchor.

- Input at reconstruction time: sparse RGB views plus known camera poses/intrinsics.
- MV-DUSt3R+ still produces the candidate point cloud.
- The fine-tuned depth model estimates per-view source depth from RGB.
- Estimated depth is back-projected and used for Run 30-style source-ray correction.

Evaluation rule

This is a target setting, not a completed claim. It must be evaluated against MV-DUSt3R+ RGB-only, generic predicted-depth correction, and the Run 30 true-depth reference.

Final Deliverables

- Reproducible Kaggle experiment scripts under `scripts/kaggle`.
- Reconstruction metrics, hard-subset gates, and qualitative 3D visualizations.
- Run 11: strongest RGB-only sparse-view baseline.
- Runs 12–29: negative reconstruction evidence for learned RGB-only filtering/correction.
- Run 30: validated RGB-D/source-depth correction reference.
- Run 32: direct RGB-D backprojection diagnostic and circularity check.
- Run 33: clean MV-DUST3R+ RGB-only candidate baseline.
- Run 34: side-by-side point clouds, dense GLBs, and normalized ND/DAC metrics.
- Run 37: fine-tuned estimated-depth source for the next correction run.
- Updated reports and slides for the RGB-only estimated-depth target setting.

Final Contributions

- A strong RGB-only baseline that solves sparse-view instability through view selection and fixed confidence thresholds.
- A negative-analysis record showing why OARH/RSDH/RAJAH and monodepth should not be claimed as RGB-only solutions.
- A validated RGB-D/source-depth reference that passes reconstruction gates for overall quality, occlusion, and repeated/wrong-depth ambiguity.
- A target RGB-only estimated-depth setting: MV-DUS_t3R+ candidates are corrected using depth predicted by the fine-tuned depth model.
- A Run 33 RGB-only backbone baseline confirming that Run 30's gain is not just MV-DUS_t3R+ candidate retention.
- A clear limitation boundary: Run 30 uses true source depth; the estimated-depth target needs separate reconstruction-gate evidence.

- Wang et al., “DUST3R: Geometric 3D Vision Made Easy,” CVPR 2024.
- Tang et al., “MV-DUST3R+: Single-Stage Scene Reconstruction from Sparse Views In 2 Seconds,” CVPR 2025.
- Yeshwanth et al., “ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes,” ICCV 2023.
- DTU Robot Image Data Sets, Multi-View Stereo Benchmark.